

How the University of Nebraska (mis)used statistics to eliminate its Department of Statistics

Authors

Department of Statistics
University of Nebraska-Lincoln

Abstract

On December 5, 2025, the University of Nebraska Board of Regents voted 7-1 to eliminate the University of Nebraska-Lincoln's Department of Statistics as part of a \$27.5M budget cut. The vote concluded the department's three-month long fight for its survival. How did this happen? The purpose of our paper is to describe the conditions that led to the original proposal for elimination and to examine the department's efforts to remain. University administrators touted their "data-informed approach" to justify the elimination of multiple departments. However, Statistics faculty discovered numerous problems with the data used and the statistical analysis performed by these administrators. Further analyses performed by the faculty showed that Statistics (and other departments proposed for elimination) actually ranked as top 20 departments at the university, rather than "lower performing" as characterized by the administration. By examining what occurred at the University of Nebraska-Lincoln, we hope this paper will help other departments—Statistics or those representing other disciplines—prevent the same from happening to them.

Keywords: Educational administration; Metric; Outlier; Program discontinuance; Shared governance; Z-score

1. Introduction

One of the most difficult decisions for university administrators is resource allocation. This requires comparing fundamentally different departments—like Statistics and Physics—to allocate resources among them. Better departments may obtain more resources than others, especially when budgets are tight. This role of administrators is much like the role of judges at dog shows when determining which dog is “best in show.” Dogs that make it to this part of the competition are from fundamentally different breeds, like Giant Schnauzer and Pug. Each breed has unique characteristics, such as size, corresponding to its purpose for being developed. How do judges compare dogs from fundamentally different breeds to determine which is best in show? The answer is these dogs are judged relative to other dogs in their *own* breed, as set by breed standards. The dog that is closest to this breed standard is awarded best in show.

University of Nebraska-Lincoln (UNL) administrators decided to make these types of comparisons among its organizational units (we will refer to these units as “departments”) in 2025. Rather than allocating resources, the purpose was to eliminate entire departments, which they believed would reduce the university’s structural budget deficit. Administrators expected that tenured/tenure-track faculty and staff would lose their jobs and students would lose their majors. These administrators were faced essentially with deciding “worst in show” among departments. The UNL Metric Analysis, a statistical analysis on data summarizing the research and instructional aspects of each department, was the focus of administrators in their decisions. Of course, the accuracy of any statistical analysis rests first on the quality of the data. Bad data will invariably result in bad decisions. If one can obtain accurate data, then appropriate statistical analyses are a must. Bad statistical analyses will invariably result in bad decisions.

The purpose of our paper is to discuss the bad data and bad statistical analyses that went into the UNL Metric Analysis. The final outcome was the University of Nebraska Board of Regents voting overwhelmingly to eliminate the following departments (projected savings included):

- Statistics, \$1.75M
- Earth and Atmospheric Sciences (EAS), \$1.85M
- Educational Administration (EDAD), \$1.69M
- Textiles, Merchandising, and Fashion Design, \$1.45M

The eliminations totaled \$6.74M of the \$27.5M in budget cuts that were approved (UNL 2025*b*). These cuts were in addition to the \$75M of other budget cuts implemented during the previous five years (UNL 2026*a*).

How did UNL get into this position? Why were these departments chosen? How can a tenured faculty member lose their job? How could a comprehensive, R1, land-grant university eliminate its Department of Statistics in the era of data science and artificial intelligence? What can be done to prevent this from occurring elsewhere? We will answer these questions in this paper. Section 2 outlines why Statistics and UNL have been in a vulnerable position for some time. Section 3 examines the proposal to eliminate Statistics, how tenured/tenure-track faculty can lose their jobs, and what will replace Statistics. Section 4 examines the UNL Metric Analysis that was touted by administrators as a “data-informed approach” for their decisions. We will present the problems with their data and statistical analysis, along with the correct way to perform the analysis. Section 5 discusses the role of shared governance during this time period. In short, all shared governance groups voted overwhelmingly to retain Statistics and highlighted problems with the UNL Metric Analysis. Section 6 examines what happened at the Board of Regents meeting on December 5, 2025, that finalized the elimination of Statistics. Section 7 concludes with events occurring since the Board of Regents meeting. This section also provides recommendations for how to prevent this from happening elsewhere. Throughout our paper, all dates refer to 2025 unless otherwise denoted.

2. Vulnerable

2.1. Statistics

While most major research universities began Departments of Statistics in the 20th century, UNL did not despite multiple attempts to form one. Instead, a “distributed model” for statistics existed, where small pockets of faculty taught and used statistics across departments. This approach was not working. Walt Stroup, the founding chair of Statistics, said in his September 29th letter to the Academic Planning Committee (APC; a shared governance committee discussed in Section 5):

By 2001 it was clear that something needed to be done. ... Richard Hoffmann arrived from Iowa State to become Dean of Arts and Sciences. His first comment when we were introduced was, “Why does UNL, a land-grant, AAU, R1 research university, not have a statistics department?”

The Department of Statistics was created on July 1, 2003, through the merger of the Statistics Division of the Department of Mathematics and Statistics in the College of Arts and Sciences (CAS) and the Department of Biometry in the Institute of Agriculture and Natural Resources (IANR; similar to a College of Agriculture). These groups were the only ones at UNL that had faculty with PhDs in statistics. This merger also combined the MS and PhD programs from each group. With its first new hire beginning in August 2003, the department started with 9.25 tenured/tenure-track faculty, and a commitment from administrators to grow the department to 13.25 total faculty.

In 2018, IANR administrators said that they wanted to own the whole department, so CAS faculty had their appointments relocated to IANR. A major change that IANR administrators wanted was for the department to start an undergraduate program. While Statistics faculty discussed starting one in the past, the resources (e.g., faculty, advisors, and space) were not available. These administrators told the faculty not to indicate in the program proposal that more resources would be needed, because this would lead to the proposal’s rejection at higher levels of the approval process. While this greatly worried many faculty, these same administrators repeatedly said “we’ve got your back” and “resources

will come.”

A Statistics faculty committee created the undergraduate program, Statistics and Data Analytics (SDAN), where “Analytics” was used in place of “Science” because administrators would not allow the latter (multiple UNL departments claim data science as their own). The program represented a reimaging of a traditional statistics undergraduate program by integrating theory and methods together into courses, teaching “soft skills” like communication and collaboration, and ensuring that students had meaningful experiences with data from their very first semester. The program was approved in fall 2020 and accepted its first cohort of students in fall 2022.

The program was lauded by others. An Academic Program Review (APR) performed by faculty from four Big 10 universities stated in their 2021 report:

The undergraduate program plan is an outstanding one—we are impressed by the vision and effort that has gone into this already. We think this is very important for UNL and the State of Nebraska, given the enormous demand for people who are highly trained in statistics. Based on what has been observed nationwide, we expect this program to grow very quickly—potentially to hundreds of majors in the near future.

The program required the creation of 24 new statistics courses. Students were required to take at least 51 credit hours from Statistics, with all of these courses taught by Statistics faculty. Figure 1 shows how this program compared to others at the time of its proposal. UNL was different in terms of the number of statistics courses and the number of faculty. It was hoped the former would help the major stand out among potential students, while also providing superior training in comparison to other statistics departments. Before the program began, the department estimated that seven new Statistics tenured/tenure-track faculty (1-2 teaching load) were needed to offer the program comfortably. Less faculty would be needed by hiring teaching-focused faculty. In fall 2023, Statistics also launched a Data Science (DS) undergraduate program with the Department of Mathematics and the School of Computing. This program incorporated some of the SDAN courses, so it did not require the department to develop any new ones.

In spring 2024, the number of Statistics faculty was 13.4, which was at approximately

the same level as when the SDAN program was approved. Faculty were feeling the strain as many were developing one or two new courses each year in addition to having new course preparations. Our outlook though remained positive. The department successfully completed an external hire that semester of a new department chair, who was promised four new faculty positions by IANR administrators.

By August 2025, the number of Statistics faculty remained at 13.4, because no hiring was allowed the previous academic year. The department was having an extremely difficult time offering the BS, MS, and PhD programs. In total, Statistics had 40 undergraduate and 42 graduate students. The 40 undergraduates may seem small in comparison to what the APR report foresaw. However, this was a result of expected resources not being delivered. A two-year, online advertising campaign targeting students alone (not high school teachers or UNL academic advisors) by IANR was insufficient. The previously mentioned strains on Statistics faculty limited our own ability to advertise as well.

2.2. UNL

Big goals came with UNL joining the Big 10 in 2011. Administrators wanted UNL to look like other Big 10 universities, with increasing student enrollment a top priority. UNL was by far the smallest public university in the Big 10 with 24,593 students (UNL 2026*b*). Chancellor Ronnie Green said during his 2017 State of the University address (Green 2017):

A reasonable 2025 goal for total enrollment for us is in the order of 29,000 to 31,000 with an upper-range, aspirational goal of 32,500 students.

Administrators had a choice similar to the classic “which came first, the chicken or the egg” problem: which comes first, the faculty or the students. IANR administrators chose faculty, so they started hiring new faculty.

UNL is the flagship university of four separate ones within the University of Nebraska (UN) System. The system hired Ted Carter to be its new president in 2020. During Carter’s four years as president, he implemented a number of changes that at first appeared to be good for UNL. We discuss three changes here. First, the Nebraska Promise allowed students to attend a university in the system for free if their family income was less than \$65K per year. Second, many faculty salaries were increased to narrow the gap between UNL

and peer universities. For Statistics, this meant some faculty received approximately 30% raises over a one-year period. Third, tuition increases were held in check. For example, in-state undergraduate tuition was raised from \$252 to \$268 per credit hour during Carter’s presidency, although the introduction of differentiated tuition rates allowed some colleges to start charging a higher value. It was often promoted by UNL Communications & Marketing that the university had the lowest tuition rate of any Big 10 university (UN System 2020). Carter left Nebraska to become Ohio State University’s president on January 1, 2024 (Smyth 2023, Smyth 2026).

UNL enrollment did not meet the big goals set by administrators. Enrollment was down to 23,954 in fall 2025, while only achieving a high of 26,079 (2017) since joining the Big 10 (UNL 2026*b*). How did UNL hire more faculty and implement Carter’s changes without substantially increasing revenue from student enrollment? This is where the \$75M in budget reductions from 2020 to 2024 (Section 1) played a significant role.

In spring 2025, the UN System requested an *increase* of 3.5% in its funding from the State of Nebraska for 2025-7. The Nebraska governor countered with a proposed 2.0% *decrease* in funding. They compromised on an increase of 1.25% (Dunker 2025). Unfortunately, the funding increase was not enough, so Chancellor Rodney Bennett started another round of budget cuts (Bennett 2025). Bennett gave the job to develop these cuts to his Executive Leadership Team (ELT); members are listed in Table 1. This team eventually developed what became known as the “Chancellor’s Proposal” for the cuts. Four of the six members of the ELT had something in common: they were interim, with Mark Button made permanent on September 10. Another member, Josh Davis, announced on November 21 that he was leaving UNL for a similar role at another Big 10 university (UNL 2025*a*).

3. Chancellor’s Proposal

On September 11, Tiffany Heng-Moss (ELT member) read the following statement during a meeting with Statistics faculty:

And so, the Department of Statistics, based upon these quantitative metrics and qualitative considerations, has been proposed for elimination.

Statistics faculty were told that this meant all tenured and tenure-track faculty would lose their jobs if the proposal was approved by the Board of Regents on December 5. Until then, faculty would have an opportunity to show why this proposal should not be adopted at a hearing with the APC (this took place on October 10).

Later in the same meeting, Heng-Moss also said the following:

I don't know how you are a comprehensive R1 land-grant institution without having a presence in statistics.

Derek McLean, Dean/Director of the Agricultural Research Division, followed Heng-Moss at this meeting by saying

I struggle. As Tiffany said, [with respect to] teaching and learning how a land-grant R1 university can then be a comprehensive university and succeed in its mission without statistical expertise.

While faculty wholeheartedly agreed with the last two quotes, it was unclear why the ELT proposed to eliminate the department. The external department chair hire with the promise of four new faculty positions seemed to show confidence in the department's future.

On September 12, UNL made the Chancellor's Proposal public (UNL 2025c). The proposal included eliminating the four departments stated in Section 1 as well as the Landscape Architecture and the Community and Regional Planning programs. This public announcement was when it was first stated that the department would be replaced with a distributed model, which was the same formulation of statistics that did not work in the past (Section 2.1). In this implementation, only five statistics courses would remain (see Table 2). Three of them were introductory statistics courses. The remaining two courses were experimental design and regression analysis at the graduate level. Additional courses with statistics content could be taught by other departments, but no specifics were provided. The Statistical Cross-Disciplinary Collaboration and Consulting Lab would remain as well. As the best Statistics faculty could determine, potentially only one faculty member from the department would remain to handle all of the consulting that was currently done by multiple faculty and graduate students.

It may seem surprising to many readers that tenured faculty could lose their jobs in this manner. After all, job security is one of the most alluring aspects of academia, and tenure

protections are an essential component of teaching and research. The Board of Regents bylaws (UN System 2025a) state four ways that tenured faculty can lose their jobs:

A Continuous Appointment may be terminated only for adequate cause, retirement for age or disability, bona fide discontinuance of a program or department, or extraordinary circumstances because of financial exigency.

Despite the structural budget deficit, the Board of Regents did not declare financial exigency. Instead, the ELT leveraged the “bona fide discontinuance of a department” to eliminate tenured faculty positions. This strategy had been used by administrators in the past to eliminate tenured faculty. The most recent attempt was in 2020 (Dunker 2020), and their last successful attempt was in 2003 (Wilson 2003; UN System 2003).

The discontinuance clause can be found in the bylaws of other university governing boards (e.g., see Kansas Board of Regents 2026 and Utah System of Higher Education 2026). Its general purpose is for situations where a major no longer attracts students or when continuing a program is not possible if most faculty leave a department. This was not the case for any of the six departments/programs slated for elimination. Many of these departments/programs offered the only degree programs of their type in the state. In particular, Statistics offered the only BS, MS, and PhD statistics programs in Nebraska.

How does one make the discontinuance bona fide then? This was the purpose of the “quantitative metrics and qualitative considerations” statement read by Heng-Moss during the September 11 meeting. The qualitative considerations were never explicitly provided to Statistics, despite UNL procedures requiring all information be made available (UNL 2022). The ELT used the UNL Metric Analysis for the quantitative metrics portion. As will be explained next, this analysis was their only attempt to justify elimination.

4. UNL Metric Analysis

Once the proposed elimination was announced on September 11, Statistics faculty requested the full data and analysis code that went into the UNL Metric Analysis. The ELT refused to release this information, so their work was not reproducible. Through leaked data and granted access to the raw research data available from Academic Analytics (AcA; this

company provided the research data to the ELT), Statistics faculty soon discovered the bad data and the bad statistical analyses that led the ELT to label departments as “lower performing.” From September 16 to October 10, faculty communicated with the ELT on 15 different days through e-mails, meetings, and a presentation to explain the problems. The ELT did not believe there were problems. On October 3, 10, 16, and 20, faculty requested that the ELT obtain an external, independent review by PhD-level statisticians. The ELT would not respond directly with an answer. On October 30, faculty invited the ELT to participate in a campuswide seminar so that both sides could present their cases. The ELT declined to participate. On November 6, Statistics faculty presented the campuswide seminar “The Metrics” to approximately 200 members of the UNL community.

We next present some of the major problems with the UNL Metric Analysis. Additional major problems are given in Web Appendix A. A detailed account of the entire process is available at SaveOurStats.com, including a recording of “The Metrics” seminar. The website also includes a report submitted to the APC (Section 5); the materials and videos for two other seminars; video testimony by faculty, students, family, and friends of the department at Board of Regents meetings; and a timeline that includes the extensive media coverage of events.

4.1. Data

A common theme among departments was

The data are not correct.

Examples provided by other departments in a public forum include those by Abrica & Garcia (2025, time 14:40) for EDAD and by Rowe et al. (2025, time 40:08) for EAS. Statistics faculty began by checking their research contributions with AcA. Table 3 shows the contribution counts for Chris Bilder, professor at UNL since 2003. One can see simply that there are differences between AcA’s counts and the actual counts for some categories. The easiest of these differences to explain involves the books count: AcA had the publication year for Bilder & Loughin (2014) recorded as 2013 rather than the actual year.

The ELT’s data missed many types of research contributions that are important to Statistics faculty at any university. Especially important are grants with faculty as a co-

investigator, consultant, or other significant contributor. Some Statistics faculty specialize in these contributions, rather than become principal investigators (PIs), because statistical help is widely needed by other disciplines. The ELT credited faculty with a grant only if they were a PI. Statistics faculty are also involved in writing software and creating apps (e.g., Hofmann et al. 2022 and Hofmann et al. 2024). In particular, Bilder created a Shiny App (Hitt et al. 2020) that helped make SARS-CoV-2 testing more efficient during the COVID-19 pandemic. This app was used by individuals in 91 countries and 47 US states during a 6-month period of 2020, and was highlighted by UNL Communications & Marketing (Schrage 2020). These types of contributions were omitted by the ELT.

The most glaring omission of a contribution involves Erin Blankenship, professor at UNL since 1999. Blankenship was awarded the University of Nebraska System’s Outstanding Teaching and Instructional Creativity Award in 2017. The President’s Excellence Awards website (UN System 2026a) says relative to this and other awards listed there that

The university’s most prestigious awards for teaching, research and engagement recognize individual faculty members and academic/administrative units whose work has had a significant impact on students, the university and the state.

These awards were deemed by the ELT as not important enough to include in their data.

The ELT excluded many Statistics faculty members from their data. Figure 2 shows the department’s faculty from 2014 to 2023, which was a time period used for some research metrics calculated upon the data. The ELT included only tenured/tenure-track faculty employed in 2025 and their corresponding contributions while at UNL during the time period. This means that any faculty member who left the department, either for another job or retirement, and non-tenure track faculty were excluded. Furthermore, the three Full Professors that joined Statistics in 2024 did not have any of their research contributions included, despite their backgrounds being greatly instrumental in their hiring. While the ELT omitted contributions by a significant number of faculty, the ELT still assumed the number of faculty during this entire time period was the same as it was in 2025.

Putting Figure 2 into numbers leads to the ELT view of Statistics as having 13 faculty for the 10-year period. This results in 130 years of potential faculty contributions to the data. However, the ELT only counted contributions from 61 of those years (47%).

The ELT also did not adjust for some faculty having administrative appointments that naturally lead to fewer research contributions. Removing those faculty (Bertrand Clarke and Blankenship) time periods results in only 45 years (35%) of potential faculty research contributions. The final outcome was the ELT including much fewer contributions than what actually occurred.

The ELT's books metric resulting from the data provides an important illustration of the problem. The ELT counted only Clarke & Clarke (2018) and attributed it to its first author, B. Clarke, alone. There were actually five books written by Statistics faculty during this time period, where the other four were

- Bilder & Loughin (2014) – Excluded because of the incorrect publication year.
- Stroup et al. (2018) and Gbur et al. (2020) – Excluded because co-author Walt Stroup retired during the time period.
- Vanderplas (2020) – Excluded because it is an online book only.

The second author of Clarke & Clarke (2018) is Jennifer Clarke, who is also a Statistics faculty member. The ELT were not aware that she was in Statistics until faculty told them, but no corrections were made for this omission (see Web Appendix A).

4.2. Z-scores

The formulation, calculation, and interpretation of z-scores are taught in introductory and multivariate statistics courses at a variety of levels. Textbooks for these types of courses include Ott & Longnecker (2016) and Johnson (1998). Because z-scores played a fundamental role in the UNL Metric Analysis, we review them here before identifying the problems with their use by the ELT.

If X is a random variable with mean μ and standard deviation σ , then a z-score is $Z = (X - \mu)/\sigma$. A z-score represents the number of standard deviations that X is from its mean. In ideal situations, such as when X has a normal distribution, the probability of Z being negative is 0.5, which in turn results in a probability of 0.5 for Z being positive. Z-scores are especially useful for identifying outliers. There are multiple definitions of outliers but all focus on Z at least being outside of ± 2 . A more conservative choice is $|Z| > 3$.

These definitions come about through using a standard normal probability distribution and Chebyshev's Theorem.

Z-scores are also useful for making comparisons among observational units with distributional differences. For instance, high temperatures during heat waves in Anchorage, Alaska, and Houston, Texas, would be quite different because of their different temperature distributions. It is not appropriate to compare these high temperatures directly to determine which city is experiencing the worst heat wave. Instead, one can transform temperatures to z-scores by estimating μ and σ from each city's historical temperature information to compare the cities in an appropriate context. We will discuss metric comparisons among departments using z-scores and alternatives in Section 4.2.3.

The ELT calculated 16 different metrics on their data for each department and transformed these metrics to z-scores. The z-scores provided to Statistics by the ELT are given in Table 4. Nine of the 16 z-scores are negative, although the ELT always stated to shared governance groups and to the Board of Regents that there were eight negative values. The number of negative z-scores here are in line with what one would expect: 50% positive and 50% negative. No z-scores are outside of ± 2 , so there are no outliers. The standard interpretation then is nothing unusual is found from these z-scores. In particular, there is nothing here that suggests that Statistics is lower performing and should be eliminated.

4.2.1. Negative values

Jason Casey (UNL Director of Institutional Effectiveness and Analytics) said the following on page 1 of a May 17 document provided by the ELT to all UNL department leaders (Web Appendix B; italicization is in the document):

Neither positive nor negative z-scores carry any interpretation other than *distance from the mean in standard deviations*. A negative score does not imply that a unit does something poorly, nor does a positive imply the opposite interpretation.

Despite the above quote, the ELT used the number of negative z-score values to justify elimination. This focus did not become clear to Statistics faculty until the APC began having hearings about the Chancellor's Proposal in early October. During these hearings,

each department gave a 50-minute presentation to appeal their recommended elimination. Before each presentation, a statement was read into the record by Josh Davis (ELT member), similar to how criminal charges are read into the record at the beginning of a court trial, that focused on negative z-score values. For example, this is what Davis said for EDAD (Abrica & Garcia 2025, time 2:52):

Educational Administration is one of the lower performing units by UNL metric analysis for instruction and research. The unit was negative on seven of nine instruction metrics and six of eight research metrics. This data has been provided to the unit.

The “metrics” are referring to the z-scores. This same type of statement was read by Davis for the Statistics hearing, but the number of negative z-scores was left out.

The ELT provided a one-page document in October to the UNL Graduate Council and to the UN System Executive Graduate Council (Web Appendix C) that made similar statements. These materials were meant to help these shared governance groups decide on how to vote regarding the proposed elimination of the departments’ graduate programs. Below is the exact entry for Statistics as it appears in the document:

Instruction: Negative on 4 of 9 Instructional metrics with a null for Retention

Instruction Outliers: Completions Majors (-1.38)

Research: negative z-scores across 4 of the 7 research metrics but no outlier more than -1

Research Outliers: N/A

The materials for the December 5 Board of Regents meeting contained the following statement regarding Statistics:

Statistics is one of the lower performing units by UNL metrics analyses for instruction and research. The department had negative scores on 4 out of 9 instructional metrics, and negative scores on 4 out of 7 research metrics.

The above are the only public statements given to the APC, the graduate councils, and the Board of Regents to justify elimination. No additional information was provided to Statistics to justify its elimination.

A larger examination of z-scores across departments illustrates further the problems with these interpretations. The ELT’s awards metric accounted for National Research Council highly prestigious awards (AAU 2021) only, rather than the broader but still limited set of awards available from AcA. Figure 3 (upper) provides a histogram of the corresponding z-scores for all departments. There were only nine unique values for these z-scores, but most troubling is that 89% of the z-scores were less than 0. Based on the ELT’s criteria for defining “lower performing,” this means the ELT believed that 89% of departments were “lower performing” relative to this metric. The same problem occurred with other metrics as well, including growth in grants at 78% and books at 71%.

4.2.2. Outliers and the completion majors metric

In addition to the ELT creating its own meaning for a negative z-score, they created their own definition for outliers to justify elimination. Again from the document provided to the graduate councils (Web Appendix C), their new definition was

Note: Outlier for this purpose defined as $+/-1$ z-score – may not align with a statistical definition.

When a z-score is applied in an ideal situation (say, the z-score has a standard normal probability distribution), the probability of being outside of ± 1 is 0.32. Thus, it is not unusual for a value to be outside of ± 1 because approximately one-third of all values would be. Even in general situations, it is not unusual for a value to be outside of ± 1 , as given by Chebyshev’s Theorem.

The ELT used the “completion majors” metric to justify eliminating Statistics. Its z-score value of -1.38 was flagged by the ELT as an outlier (see quoted text in Section 4.2.1) using their new definition. The formulation of the metric used the ratio (number of graduations)/(number of majors) for the 2023-4 year, which was further transformed to a z-score. The numerical values used in the ratio were across the department’s undergraduate and graduate programs without any distinction between them. The very likely reason why this z-score was negative is because the department started its first undergraduate program in 2022 (SDAN) and its second in 2023 (DS). The undergraduate students did not graduate yet because these degrees take four years to complete, like most

bachelor’s degree programs in any field. Therefore, the denominator in the ratio was much larger than the numerator because SDAN and DS graduations were not possible, which in turn resulted in the -1.38 value. This metric should have been thrown out of the UNL Metric Analysis or adjusted to reflect only the department’s graduate programs.

The ELT was aware of the new undergraduate programs. When Statistics faculty questioned them about why they were reporting the z-score for completion majors, Mark Button (ELT member) indicated that

Major completions is one score out of 18 [sic] quantitative assessments, and we did consider the important fact that Statistics began offering an undergraduate major in 2022.

Unfortunately, this did not change that the metric was flagged as a outlier and stated to shared governance groups and the Board of Regents as one of four negative instructional z-scores. No appropriate context was provided with this metric at any time.

4.2.3. Formulation

The formulation of z-scores was the ELT’s fatal flaw in their UNL Metric Analysis. This flaw can make middle-of-the-road departments look like the best, while higher performing departments look the opposite. We provide a technical and non-technical explanation of their flaw and follow it with a solution.

Define X_i as a random variable with mean μ_i and standard deviation σ_i for $i = 1, \dots, n$. An X_i represents a specific metric for department i . For example, X_1 could represent the citations metric for Statistics, and X_n could represent the same metric for Physics and Astronomy (we refer to the department as “Physics” for brevity). We include subscript i ’s on the mean and standard deviation to allow for differences among departments. This generalization from the previously stated Z is needed because each X_i potentially has a different distribution: departments value books, citations, teaching methods, and other metrics differently.

A z-score in this setting is formed as

$$Z_i = \frac{X_i - \mu_i}{\sigma_i}.$$

The scenario faced by the ELT was to compare Z_1 to Z_2 to ... to Z_n . Relative to the previous example, Z_1 could represent the z-score of the citations metric for Statistics, and Z_n could represent the same for Physics. The ELT viewed $Z_n > Z_1$ as indicating Physics was better than Statistics for a particular metric.

The ELT's fatal flaw was to estimate μ_i by the overall mean of the observed values for $X_1, \dots, X_n$. A similar mistake was made by estimating σ_i with the overall standard deviation as well. These estimates are incorrect because each X_i potentially has a different mean and a different standard deviation. Comparisons of Z_1 to Z_2 to ... to Z_n are nonsensical. We demonstrate this later with the citations metric comparison of Statistics to Physics.

The non-technical explanation of the problem involves the previous best in show example from Section 1. To help relate this to the ELT's fatal flaw, consider the best in show portion of the National Dog Show in 2024. Figure 4 shows the dogs considered. A judge compared each dog to the standards of its own breed to decide which dog was best. Thus, Monte the Giant Schnauzer, the far-left dog in Figure 4, was compared against other Giant Schnauzers. The dog closest to its standard was awarded best in show.

Suppose instead that one ignored these dogs are from fundamentally different breeds. And, suppose size, where the larger the better, is the only metric used to compare dogs. One would choose Monte the Giant Schnauzer then as best in show, because Monte is the largest among the other dogs. This is essentially what the ELT did when comparing departments. They ignored the fundamental differences among departments in forming their z-scores. As discussed earlier, the ELT tried to find the worst in show among departments rather than best in show. Applying their methodology, Vito the Pug dog would be eliminated, because it is the smallest dog in Figure 4. Again, this incorrect approach ignores the fundamental differences among breeds: Pugs are naturally small dogs.

The correct approach for comparing departments is to estimate μ_i and σ_i by a sample from within each discipline. For example, one could estimate the mean for Statistics citations by obtaining the citations values for other Departments of Statistics and average them. However, rather than examining this approach, we focus on a better and easier solution that allows one to compare UNL departments using ranks within discipline. Below is the two-step process:

1. Find the percentile ranking of each UNL department within its discipline across universities.
2. Compare these percentiles for each UNL department.

To illustrate this process, we examine the Scholarly Research Index (SRI; see Web Appendix A) metric used by the ELT. This is a metric created by AcA to measure overall research performance by a department. UNL Statistics is at the 75th percentile (39th best out of 158 departments within the statistics discipline) for the SRI. To further examine this ranking, Figure 5 provides plots of the SRI values for R1 universities, where we condition on Association of American Universities (AAU) membership (all AAU institutions are R1). This AAU split is especially important because it is a priority for administrators and Regents to get UNL back into the AAU after being expelled in 2011 (Abourezk 2011, Clare 2023, Dunker 2023, Moody 2023). The plot shows that UNL Statistics is comfortably in the middle of the AAU institutions. It makes no sense to eliminate a department that actually would help UNL look like an AAU institution.

Figure 6 displays the percentile rankings of all UNL departments. Statistics is a top-20 UNL department with a ranking of 17th. This is quite different from the ELT representation of Statistics as “lower performing” that was stated in Section 4.2.1. Furthermore, the plot shows that the ELT originally wanted to eliminate four of UNL’s best 17 research departments. This illustrates just how flawed the ELT’s statistical analysis was in practice.

To explore the ELT’s fatal flaw further, we compare Statistics to Physics using the citations metric. Figure 3 (lower) provides a histogram of the z-score values for all departments with Statistics and Physics denoted. Physics has a positive value much larger than all other departments, indicating this department is by far the best relative to the metric and the ELT’s interpretations. On the other hand, Statistics has a negative z-score value, so it is considered “lower performing” by the ELT.

When the departments are compared the correct way through percentiles relative to their own discipline, there is a dramatic change in understanding which department is better. Statistics is at the 79th percentile, and Physics is at the 40th percentile. Thus, Statistics is a top quartile department when compared to its own discipline, while Physics is a middle-of-the-road department relative to its own discipline. This change occurs because

of differences in the citation distributions for the two disciplines. The culture of the physics discipline leads to a very large number of citations in comparison to others.

5. Shared Governance

Shared governance groups voted in October on whether to eliminate Statistics and its degree programs. The outcomes were as follows:

- UNL Graduate Council (2025) voted 10-0 to retain the graduate programs.
- UN System Executive Graduate Council (2025) voted 8-0 to retain the graduate programs.
- APC (2025) voted 13 to 8 to retain Statistics.

All proposed-to-be-eliminated programs had the same vote totals for the graduate councils, while the APC votes were mixed. APC members were faculty (11), administrators (6), staff (2), and students (2), where 4 of the 6 administrators were also ELT members. These ELT members refused to recuse themselves from voting when asked by the faculty. Based on APC member comments, this appears to be the first time recusal did not occur from individuals who constructed elimination proposals.

Statements in reports and letters written by these shared governance groups echoed the problems described by the Statistics faculty to the ELT and urged the administration to stop the elimination process for all departments. Below is a sample of the statements from these documents, where any bolding was in the original documents.

- We found no evidence that any of the affected graduate programs meet the standards for deletion established by the Graduate College Bylaws.
- **We strongly recommend to the Chancellor, the President, and the Board of Regents that the approval of any budget cuts be delayed allowing time for units to identify creative alternative solutions that reduce or prevent the need for these cuts. . . . Moreover, to provide this delay for academic cuts costs nothing**, as the timeline for implementation of this process is at least 14 months, and much longer to accommodate teach-out plans.

- Although the metric definitions were confidentially shared with Deans and with the APC last spring, no one was able to fully validate the metrics, either through confirming the accuracy of the underlying data or via analysis to confirm that the metrics were statistically valid ways to quantify the desired performance indicators.
- There were concerns raised about the alignment of metrics with the goals and strategic plan for the University. For example, rejoining the AAU is an often-cited goal for the University, but some units that rated very highly compared to AAU peers were proposed for elimination. On the other hand, rejoining the AAU is not necessarily well-aligned with our responsibility as a land-grant university with a significant outreach mission.
- Before proceeding, an independent verification of data and methodology is essential to ensure the Regents' decision rests on accurate and defensible information.

Additional separate statements were specific to each department as well. None of the statements for Statistics supported its elimination.

The shared governance process began in August and concluded on October 31 with the release of the APC report to the public. The APC vote was expected to significantly influence the final budget reduction proposal. At no time during previously proposed eliminations had the process moved forward without APC approval.

On November 10, Chancellor Bennett released his final budget reduction proposal. The Landscape Architecture and the Community and Regional Planning programs were spared, because these programs found a larger budget savings by other means than through elimination. It is important to note that their college dean was an APC member. Statistics and the remaining three departments were still proposed to be eliminated. On November 18, the Faculty Senate voted "no confidence" in Bennett 60 to 14 with 3 abstentions. This was the first time in UNL's history that the senate voted no confidence in the Chancellor. The next and last opportunity to save Statistics was the Board of Regents. Their vote was scheduled for December 5 in an open-to-the-public meeting.

6. Board of Regents

Statistics faculty continued their efforts after the final Chancellor’s Proposal. They met with David Jackson (Interim Executive Vice President and Provost of the UN System) who said a “Hail Mary” was needed to save the department. Jackson said Statistics needed to find \$1.75M (projected savings from elimination) on its own. Statistics faculty contacted Board of Regents members, but only Regent Tim Clare (elected representative for district that includes UNL) agreed to meet with department representatives, Bilder and Blankenship. Their discussion included the problems with the UNL Metric Analysis. Clare requested Bilder’s phone number and said he would follow up with him after talking to Mark Button (ELT member), but the promised phone call never occurred. Statistics Head Bhaskar Bhattacharya developed a last-minute, \$1.75M savings/revenue plan and presented it to the Board of Regents. This plan was ignored. A detailed explanation for its rejection was not provided until March 16, 2026, after multiple requests made by the department.

There was an usually large turnout for the Board of Regents meeting on December 5. The board room, an overflow room, and the halls were full of people there to testify and to support the departments. A total of 109 people testified—all against the elimination of the departments (UN System 2025*b*, time 1:11:12). Each person was allotted three minutes, which led to more than five hours of testimony, without any breaks. The video testimonies from all 24 individuals who spoke on behalf of Statistics are available from our website (UN System 2025*c*). These testimonies describe the problems with the UNL Metric Analysis, but also detail alternative cost savings options, the violation of UNL rules by the ELT, and the overall importance of statistics to the university and the state.

We want to bring special attention to the second testimony in the video. This was given by Alex Vanderplas, the 9-year old son of Statistics faculty member Susan Vanderplas. A transcript is provided in Web Appendix D. A. Vanderplas stood before the board and described his aspirations to be a paleontologist when he grows up (EAS provides the required geology education), the importance of statistics to science, and the effect these eliminations would have on families. This testimony puts into context the devastation that our home and work families feel, especially with the irony that a misuse of statistics played a central

role in the elimination of Statistics.

After the testimonies, some Regents spoke to explain why they were going to vote for elimination (UN System 2025*b*, time 7:32:10). Clare said the following about the UNL Metric Analysis:

I appreciate the efforts and I trust the work they [UN System President Jeff Gold, Chancellor Bennett, ELT] put out, put together, while people may dispute the data, they did, that team of people did what we asked them to do and that was figure out how we can solve this potential budget shortfall.

Jim Scheer, Regent and former Speaker of the Nebraska Legislature, suggested that the supporters of the eliminated departments were partially at fault. He spoke directly to the audience:

My question to you, of the 100 people here, how many of you went to the Legislature when they were having a hearing on the university's budget. OK, less than 10. ... The Legislature is a large part of what we're doing right now. ... You got to the game late.

The Board of Regents voted 7-1 to eliminate Statistics. December 5th was a sad day for all of statistics.

7. Conclusion

We are proud of what the department accomplished in its short 23-year history. We were able to merge two diverse cultures from CAS and IANR into one department. We are most proud of our hundreds of graduates, who live around the world working in industry, government, and academics. We are also proud of the statistics community. We asked for help in September, and they delivered with approximately 500 e-mails and comments of support submitted to administrators and the APC over a one-month period. In the end, a few administrators led UNL down what we believe is the wrong path. We hope the university can recover.

More continues to occur after the Board of Regents meeting. Important events include (all dates are 2026):

- January 5: Rodney Bennett announced his resignation as Chancellor. The Board of Regents agreed to pay him \$340K for the remaining 5.5 months of salary on his contract and \$1.1M more for leaving (UN System 2026*b*). All parties signed non-disclosure agreements.
- January 15: A listening session hosted by UN System President Jeff Gold and Interim Chancellor Kathy Ankerson was held for the UNL community to quell its dissatisfaction after the eliminations (UNL 2026*c*). Gold alluded multiple times to mistakes that others made during the budget reduction process. This included the statement:

I trusted people who said that they were accurately communicating, that there were metrics that were validated and agreed upon. I trusted people that said that there was formal input on multiple levels of appropriate shared governance. I trusted that. And my trust was gone.

Gold backtracked from these statements the next day in an e-mail sent to the UNL community (Gold & Ankerson 2026).

- January 27: During a Faculty Senate Executive Committee meeting, Mark Button (ELT member) announced that there an additional \$4M saved from the voluntary separation incentive program than expected (a separate part of the overall budget reduction plan). These funds would be for “strategic faculty investment” to achieve UNL’s goal of AAU membership (Griffin 2026). Shortly after Button left the meeting, termination letters were sent to faculty of the four eliminated departments, including to faculty still at the meeting. Two Statistics faculty did not receive letters and instead are transitioning into a new School of Agricultural Social Sciences and Analytics. For the remaining Statistics faculty, employment is guaranteed until May 2027.
- February 17: Interim Dean Tom Burkey in IANR announced that a total of 12 courses from Statistics will remain, along with two new courses to be developed. These 14 courses are a significant increase over the five courses that were planned (Section 3). No additional Statistics faculty have been retained to teach these courses.

- March 30: Interim Chancellor Ankerson announced a new strategic framework for the university called “Our Bold Path Forward” that looks past what happened in fall 2025.

There is no more Department of Statistics at UNL. The worst in show was incorrectly identified through the use of bad data and bad statistical analyses and subsequently eliminated. We hope a similar outcome does not happen elsewhere for any department. To help prevent it, we provide four suggestions:

1. Get to know the institutional research office that holds the official university data. Audit their data and create formal data validation procedures through shared governance. Work with this office to ensure they have the appropriate statistical and computing skills. Require data correctness and reproducible calculations.
2. Size matters and money matters. It is much easier to eliminate small programs than large ones. The ELT claimed to account for department size, but this was demonstrated to be incorrect (see Web Appendix A). Look for ways to increase the number of majors and revenue, while also trying to be more efficient.
3. Encourage department faculty to be part of shared governance and become administrators. Statistics had three faculty who were part of shared governance. This greatly helped all faculty understand many issues regarding the proposed elimination and provided direct lines of communication with administrators, although it did not save Statistics in the end.
4. Insist on faculty experts in fields that can help, such as statistics and educational administration, take part in any budget reduction process.

Supplementary materials

Additional materials may be found in the online version of the article at the publisher’s website and at <WEBSITE>. These materials include our web appendices and an R program.

Disclosure statement

The authors report that there are no competing interests to declare.

Data availability

The data are available in our supplementary materials.

References

- AAU (2021), ‘National Research Council list of highly prestigious awards’.
<https://www.aau.edu/key-issues/national-research-council-list-highly-prestigious-awards>. Retrieved May 4, 2026.
- Abourezk, K. (2011), ‘Research universities group ends UNL’s membership’, *Lincoln Journal Star* . April 29; Retrieved May 4, 2026.
- Ablica, E. & Garcia, C. (2025), ‘Educational Administration’.
<https://mediahub.unl.edu/media/25752>. Retrieved May 4, 2026.
- APC (2025), ‘Recommendations on proposed budget reductions’.
https://budgetprocess.unl.edu/sites/unl.edu.budget-process/files/media/file/20251024_apc_recommendation.pdf. Retrieved May 4, 2026.
- Bennett, R. (2025), ‘Message from the Chancellor - budget reductions’.
<https://chancellor.unl.edu/message-chancellor-budget-reductions>. Retrieved May 4, 2026.
- Bilder, C. & Loughin, T. (2014), *Analysis of Categorical Data with R*, CRC Press.
- Clare, T. (2023), ‘Full steam ahead for NU’, *Lincoln Journal Star* . August 29; Retrieved May 4, 2026.
- Clarke, B. & Clarke, J. (2018), *Predictive Statistics: Analysis and Inference Beyond Models*, Cambridge University Press.

- Dunker, C. (2020), ‘UNL proposes cutting \$18.9 million in faculty, staff positions over three years’, *Lincoln Journal Star* . September 3; Retrieved May 4, 2026.
- Dunker, C. (2023), ‘Everything AAU - Even with president’s departure, NU still eyeing reentry to prestigious group’, *Lincoln Journal Star* . September 3; Retrieved May 4, 2026.
- Dunker, C. (2025), ‘NU president: Nebraska budget could mean tuition increases this year’, *Lincoln Journal Star* . May 6; Retrieved May 4, 2026.
- Gbur, E., Stroup, W., McCarter, K., Durham, S., Young, L., Christman, M., West, M. & Kramer, M. (2020), *Analysis of Generalized Linear Mixed Models in the Agricultural and Natural Resources Sciences*, John Wiley & Sons.
- Gold, J. & Ankerson, K. (2026), ‘A letter to the UNL campus community’. <https://chancellor.unl.edu/letter-unl-campus-community>. Retrieved May 4, 2026.
- Green, R. (2017), ‘State of the University Address 2017-2018’. <https://chancellor.unl.edu/state-university-address-2017-2018>. Retrieved May 4, 2026.
- Griffin, K. (2026), ‘Faculty Senate Executive Committee minutes’. January 27; <https://www.saveourstats.com/letters/26Jan27mins.pdf>. Retrieved May 4, 2026.
- Hitt, B., Bilder, C., Tebbs, J., McMahan, C. & Yaseen, M. (2020), ‘A Shiny App for Pooled Testing’. <https://bilder.shinyapps.io/PooledTesting>.
- Hofmann, H., Vanderplas, S. & Ge, Y. (2022), ‘ggpcp: Parallel coordinate plots in the ‘ggplot2’ framework’. <https://cran.r-project.org/package=ggpcp>. Retrieved May 4, 2026.
- Hofmann, H., Vanderplas, S., Krishnan, G. & Hare, E. (2024), ‘x3ptools: Tools for working with 3D surface measurements’. <https://cran.r-project.org/package=x3ptools>. Retrieved May 4, 2026.
- Johnson, D. (1998), *Applied Multivariate Methods for Data Analysts*, Duxbury Press.

- Kansas Board of Regents (2026), 'Kansas Board of Regents Policy Manual'. <https://www.kansasregents.gov/about/policies-by-laws-missions>. February 11 revision. Retrieved May 4, 2026.
- Moody, J. (2023), 'Nebraska seeks a return to the AAU', *Inside Higher Ed* . August 15; Retrieved May 4, 2026.
- Ott, R. & Longnecker, M. (2016), *An Introduction to Statistical Methods and Data Analysis*, Cengage Learning Inc.
- Rowe, C., Mauntel, K., Szot, R., Houston, A., Haacker, E., Lewis, B., Burberry, C. & Wagner, J. (2025), 'Earth and Atmospheric Sciences: Testimony to the Academic Planning Committee against program elimination'. <https://mediahub.unl.edu/media/25758>. Retrieved May 4, 2026.
- Schrage, S. (2020), 'Statisticians advise lab, release app to optimize coronavirus testing', *Nebraska Today* . April 21; Retrieved May 4, 2026.
- Smyth, J. (2023), 'New president of Ohio State will be Walter 'Ted' Carter Jr., a higher education and military leader', *Associated Press* . August 15; Retrieved May 4, 2026.
- Smyth, J. (2026), 'Ohio State University's president resigns after reporting 'inappropriate relationship'', *Associated Press* . March 9; Retrieved May 4, 2026.
- Stroup, W., Milliken, G., Claassen, E. & Wolfinger, R. (2018), *SAS for Mixed Models: Introduction and Basic Applications*, SAS Institute.
- UN System (2003), 'Board of Regents Meeting Minutes'. August 9; <https://mediacommons.unl.edu/luna/servlet/UNL~152~152>. Retrieved May 4, 2026.
- UN System (2020), 'University to offer free tuition to Nebraska students with family incomes of \$60,000 or less'. <https://nebraska.edu/news-and-events/news/2020/04/university-to-offer-free-tuition-to-nebraska-students-with-family-incomes-of-60000-or-less>. Retrieved May 4, 2026.

UN System (2025a), ‘Bylaws policies and rules’. <https://nebraska.edu/regents/bylaws-policies-and-rules>. Retrieved May 4, 2026.

UN System (2025b), ‘University of Nebraska Board of Regents meeting: December 5, 2025.’. <https://www.youtube.com/live/K5G-aeDMiCc>. Retrieved May 4, 2026.

UN System (2025c), ‘University of Nebraska Board of Regents Meeting on December 5, 2025; testimony by Department of Statistics’. <https://www.youtube.com/watch?v=1z3Bp5mdvG8>. Retrieved May 4, 2026.

UN System (2026a), ‘President’s Excellence Awards’. <https://nebraska.edu/president/faculty-engagement/presidents-excellence-awards>. Retrieved May 4, 2026.

UN System (2026b), ‘Resignation agreement and release’. <https://bloximages.chicago2.vip.townnews.com/journalstar.com/content/tncms/assets/v3/editorial/1/6f/16f8b8bc-23d7-4d9c-9ec5-a9401cd5bfe6/695c3c88c8a93.pdf.pdf>. Retrieved May 4, 2026.

UN System Executive Graduate Council (2025), ‘Letter to University of Nebraska Board of Regents’. <https://www.saveourstats.com/letters/EGCemailBOR.pdf>. Retrieved May 4, 2026.

UNL (2022), ‘Procedures to be invoked for significant budget reallocations and reductions’. <https://apc.unl.edu/documents-reports>. Retrieved May 4, 2026.

UNL (2025a), ‘Farewell reception for Davis is Dec. 8’, *Nebraska Today* . <https://news.unl.edu/article/farewell-reception-for-davis-is-dec-8>. Retrieved May 4, 2026.

UNL (2025b), ‘Final budget reduction plan’. <https://budgetprocess.unl.edu/final-budget-reduction-plan>. Retrieved May 4, 2026.

UNL (2025c), ‘Proposed budget reductions’. <https://budgetprocess.unl.edu/proposed-budget-reductions>. Retrieved May 4, 2026.

UNL (2026a), ‘Budget process’. <https://budgetprocess.unl.edu>. Retrieved May 4, 2026.

UNL (2026b), ‘Fact Book’. <https://iea.unl.edu/publication/fact-book>. Retrieved May 4, 2026.

UNL (2026c), ‘UNL Campus Listening Session’. <https://www.youtube.com/watch?v=ifihZmJSbLI&t=2611s>. Retrieved May 4, 2026.

UNL Graduate Council (2025), ‘Letter to University of Nebraska Executive Graduate Council’. <https://www.saveourstats.com/letters/UNLgraduateCouncil.pdf>. Retrieved May 4, 2026.

Utah System of Higher Education (2026), ‘Utah System of Higher Education Policies’. <https://public.powerdms.com/Uta7295/tree>. Retrieved May 4, 2026.

Vanderplas, S. (2020), *Statistical Computing in R and Python*. <https://srvanderplas.github.io/stat-computing-r-python>.

Wilson, R. (2003), ‘U. of Nebraska at Lincoln plans to lay off 8 tenured professors in budget cuts’, *The Chronicle of Higher Education*. March 11; Retrieved May 4, 2026.

Table 1: Executive Leadership Team members early September 2025. Names of concurrent members of the APC (Section 5) are bolded. Button became permanent in fall 2025. Moss and Nelson became permanent in spring 2026. Davis announced he was leaving for Penn State on November 21.

Name	Title
Dee Dee Anderson	Vice Chancellor, Student Life
Anne Barnes	Interim Vice Chancellor, Business and Finance
Mark Button	Interim Executive Vice Chancellor, Chief Academic Officer
Josh Davis	Vice Chancellor, Instructional Strategy and External Relations
Tiffany Heng-Moss	Interim Vice Chancellor, IANR
Jennifer Nelson	Interim Vice Chancellor, Research and Innovation

Table 2: Statistics courses for the distributed model.

Course number	Title	Level
STAT 218	Introduction to Statistics	Undergraduate
STAT 380	Statistics and Applications	Undergraduate
STAT 801	Statistical Methods in Research	Graduate
STAT 802	Design and Analysis of Research Studies	Graduate
STAT 870	Multiple Regression Analysis	Graduate

Table 3: Counts for Chris Bilder’s research contributions. The large discrepancy in citations occurs because the ELT chose not to include citations from papers that were published prior to 2019. Because of the slower publishing and citation culture of the statistics discipline, this limits understanding a paper’s impact. The data were obtained from AcA, Bilder’s curriculum vitae, and Bilder’s Google Scholar web page in September 2025.

Contribution	AcA	Correct
Publications, 2020-2023	10	14
Awards, no time limit	3	7
Books, 2014-2023	0	1
Chapters, 2020-2023	0	0
Citations, 2019-2023	353	1264
Conference proceedings, 2020-2023	0	1
Grants awarded as PI, 2019-2023	1	1

Table 4: Z-scores for Statistics as provided by the ELT to the department. Student credit hours are abbreviated as SCH, and full-time equivalent is abbreviated as FTE. The ELT assigned a “NULL” for freshman retention rather than calculate its actual value. The averages at the bottom of the table were used by the ELT to simplify their analysis. These averages are discussed in Web Appendix A.

Research	Z-score	Instructional	Z-score
Federal research expenditures	0.19	SCH growth	-0.55
Awards	-0.33	Majors growth	0.11
Books	-0.56	SCH	-0.26
Citations	-0.09	Majors	-0.89
Scholarly Research Index	0.55	SCH relative to FTE	0.74
Grants relative to state funding	-0.40	Budget relative to SCH	0.14
Growth in grants	-0.43	Tuition relative to budget	0.05
		Completion of majors	-1.38
		Freshman retention	NULL
Average	-0.15	Average	-0.26

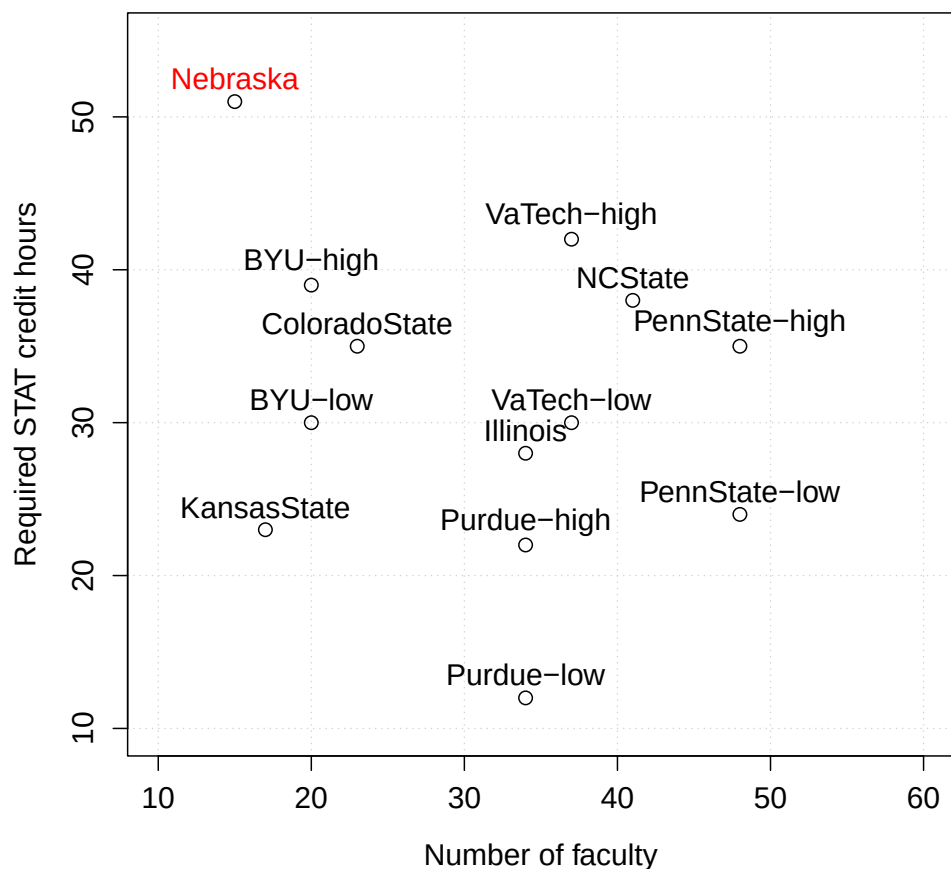


Figure 1: Required number of credit hours for a statistics undergraduate major that would likely be taught by Statistics faculty (course housed within Statistics). Universities represented here are peers rather than a random sample. The “low” and “high” representations for the same university occur because students can choose from a list of courses that may or may not be housed within Statistics. The data were collected in spring 2020.

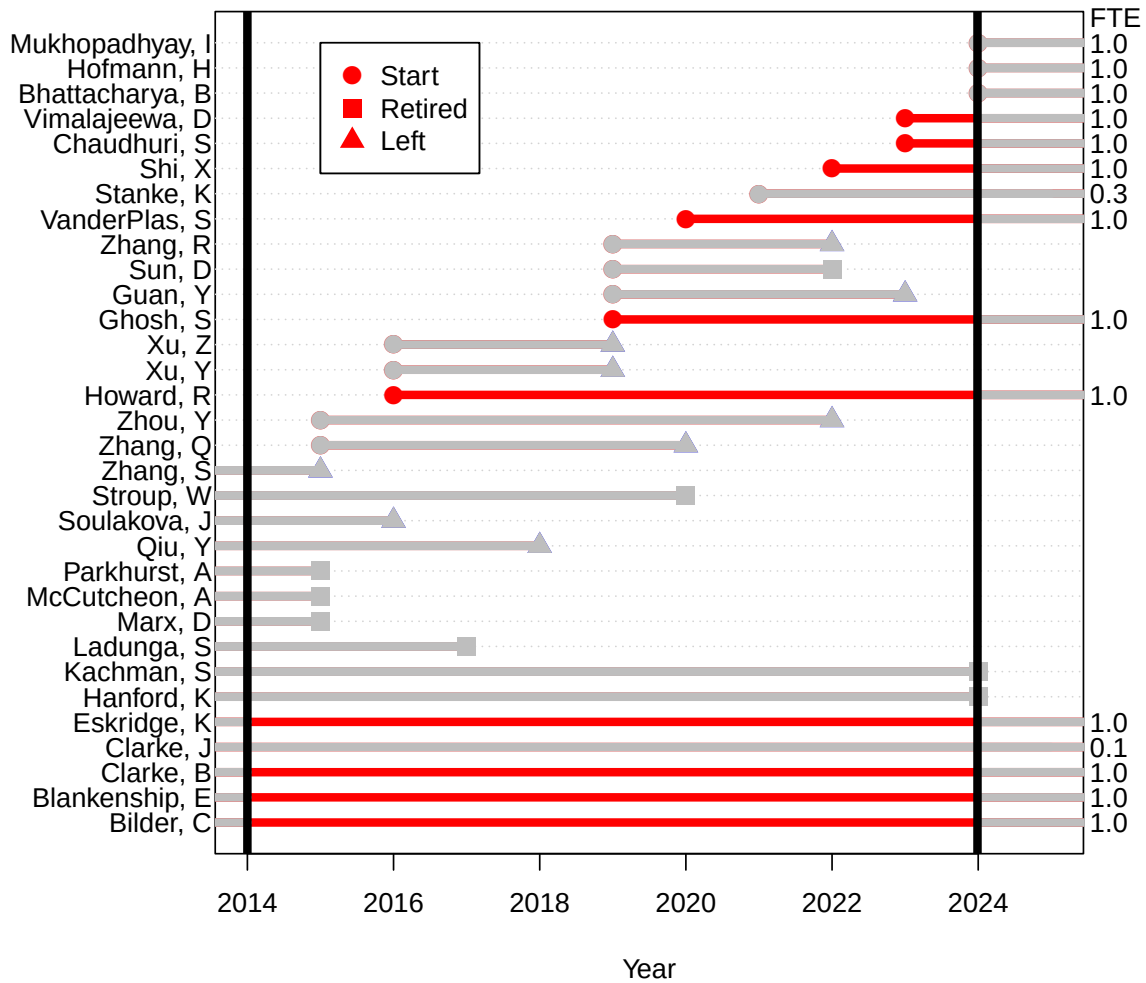


Figure 2: Faculty composition of Statistics. Thick horizontal lines indicate years that faculty were part of the department. Red lines correspond to years which research contributions were included by the ELT in their data.

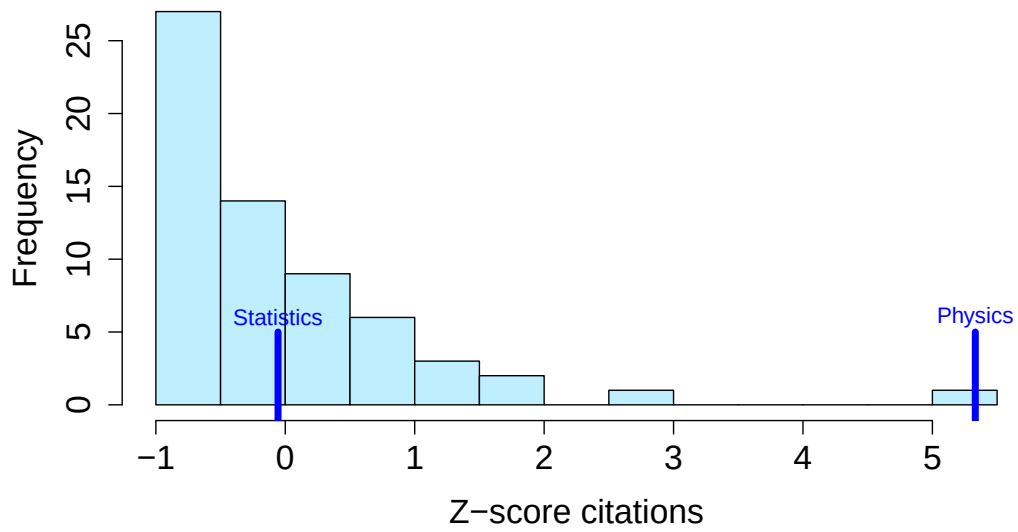
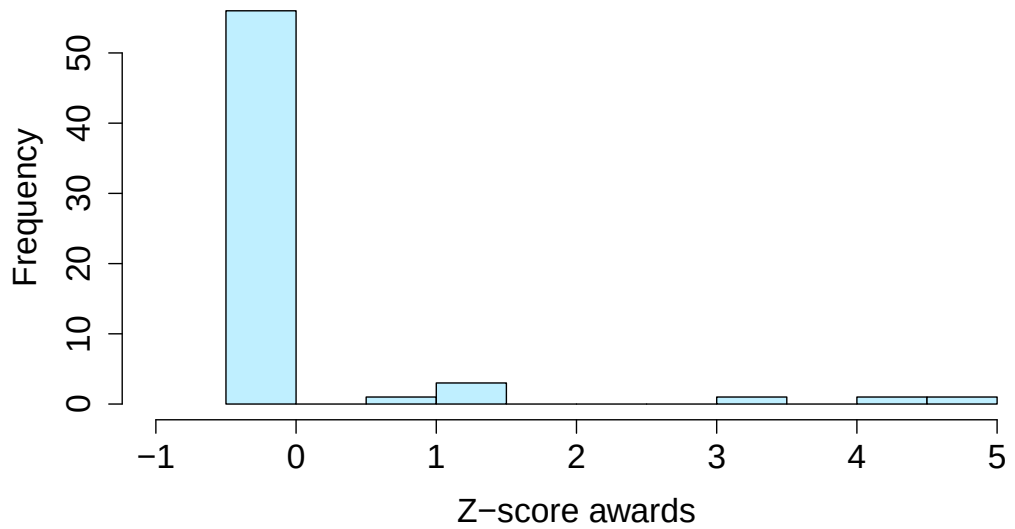


Figure 3: Histograms of departmental z-scores. Upper plot is for the awards metric. Lower plot is for the citations metric. The data were obtained from an August 29th dated document shared with the APC.



Figure 4: Best in Show competitors at the National Dog Show in 2024. From left to right: Monte the Giant Schnauzer, Rupert the Berger Picard, The Zit the Ibizan Hound, Houston the Clumber Spaniel, Verde the Welsh Terrier, JJ the Lhasa Apso, and Vito the Pug. Vito won Best in Show. Source: <https://www.youtube.com/watch?v=gpls8xguMn8&t=1120s>.

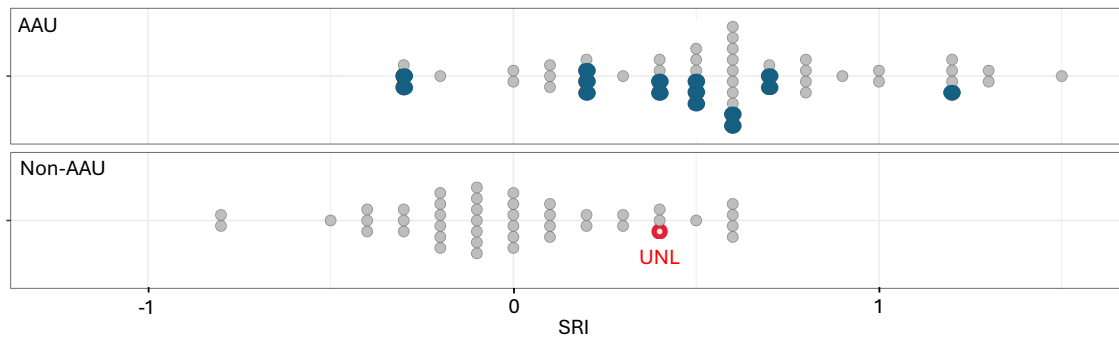


Figure 5: SRI values for Departments of Statistics in R1 universities. The blue dots represent a natural group of universities that includes UNL as a member. This group is not named here because of the licensing agreement between UNL and AcA. The data were obtained from AcA.

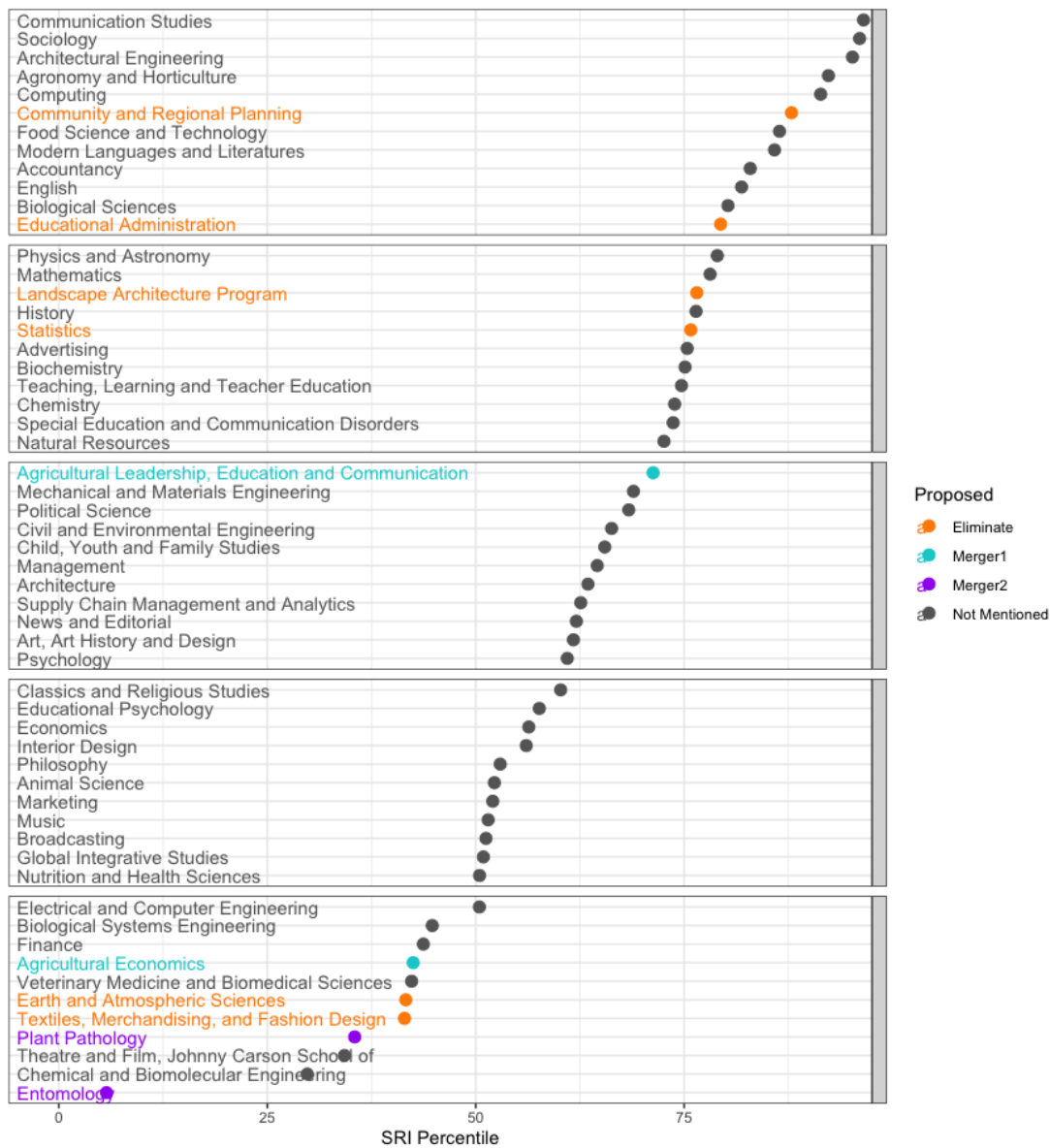


Figure 6: Percentile rankings of departments/programs with plotting point color corresponding to the original Chancellor's Proposal. Some departments/programs had multiple rankings because they contain multiple disciplines. In those cases, we represented them by their maximum ranking. It is important to note that some departments/programs, like Textiles, Merchandising, and Fashion Design, are not properly assigned to a discipline area within AcA, which greatly affects their percentile. The data were obtained from AcA.